Juarez Monteiro

AI Researcher and Data Scientist

LinkedIn | Google Scholar | YouTube | GitHub

Ph.D. in Computer Science from PUCRS. I research imitation learning, reinforcement learning, foundation models for tabular data and generative AI. I work in industry as a researcher and data scientist.

Education

MBA in Leadership, Team Management and Productivity — PUCRS, Pontifical Catholic University of Rio Grande do Sul (Jul 2026 – present)

Ph.D. in Computer Science — PUCRS, Pontifical Catholic University of Rio Grande do Sul (2018 – 2023)
Advances in Imitation Learning from Observation

M.Sc. in Computer Science — PUCRS, Pontifical Catholic University of Rio Grande do Sul (2016 – 2018)
Evaluating the feasibility of deep learning for action recognition in small datasets

Associate Degree in Systems Analysis and Development — IFRS, Federal Institute of Education, Science and Technology of Rio Grande do Sul (2011 – 2014)
Analysis of the Applicability of Artificial Intelligence Techniques in a Digital Game: An Approach with Intelligent Agents

Industry experience

AI Researcher and Data Scientist — Dell Technologies (Jul 2023 – present)

Data Scientist — Sicredi (May 2022 – Sep 2023)

ML Engineer — unico IdTech (Nov 2021 – Apr 2022)

Intern – Software Developer — CWI Software (Apr 2015 – Sep 2015)

Intern – Performance Tester — Dell Technologies (Dec 2013 – Feb 2015)

Intern – Service Reports — Stefanini IT Solutions (Mar 2013 – Jul 2013)

Teaching

Instructor – Machine and Deep Learning 1 — Feevale (2021)

Instructor – Data Science — TargetTrust Courses (2020 – 2021)

Instructor – Python for Data Science — Feevale (2018)

Teaching Assistant – Business Intelligence — PUCRS (2018)

Teaching Assistant – Database Laboratory — PUCRS (2018)

Awards

2nd best Master's dissertation in Artificial Intelligence in Brazil — PhD Thesis Contest in Artificial Intelligence, CTDIAC-BRACIS, 2018

Best Paper — IEEE IJCNN, 2017

Best Student Poster — IFRS, Canoas Campus, 2015

Skills

Machine learning: Imitation Learning, Reinforcement Learning, Deep Learning, Generative AI

Foundation models and LLMs: Tabular Foundation Models (TabPFN), Language Models, Uncertainty Estimation

Tools: Python, PyTorch, OpenAI Gym, Git

Data and healthcare: Data Science, Synthetic EHR data, Clinical concept normalization

Publications

- When to ASK: Uncertainty-Gated Language Assistance for Reinforcement Learning. **Juarez Monteiro**, Nathan Gavenski, Gianlucca Zuin, Adriano Veloso. *IJCNN 2026*, 2026.
- When in Doubt, Plan It Out: Committed Small Language Model Deliberation for Reactive Reinforcement Learning. Nathan Gavenski, **Juarez Monteiro**, Francisco Galuppo Azevedo, Adriano Veloso, Odinaldo Rodrigues. *ICML 2026 · LM4Plan Workshop*, 2026.
- DataSynK: Causal-Symbolic EHR Synthesis for Tabular Foundation Models in Low-Resource Settings. Eduarda T. C. Chagas, Roberta Viola, **Juarez Monteiro**, Francisco Galuppo Azevedo, Saulo Fernandes Saturnino, Adriano Veloso. *ICML 2026 · Foundation Models for Structured Data Workshop*, 2026.
- Self-Supervised Adversarial Imitation Learning. **Juarez Monteiro**, Nathan Gavenski, Felipe Meneguzzi, Rodrigo Barros. *IJCNN 2023*, 2023.
- How Resilient Are Imitation Learning Methods to Sub-optimal Experts?. Nathan Gavenski, **Juarez Monteiro**, Roger Granada, Adilson Medronha, Rodrigo Barros. *BRACIS 2022*, 2022.
- Augmented Behavioral Cloning from Observation. **Juarez Monteiro**, Nathan Gavenski, Roger Granada, Felipe Meneguzzi, Rodrigo Barros. *IJCNN 2020*, 2020.
- Imitating Unknown Policies via Exploration. Nathan Gavenski, **Juarez Monteiro**, Roger Granada, Felipe Meneguzzi, Rodrigo C. Barros. *BMVC 2020*, 2020.
- Object-Based Goal Recognition using Real-World Data. Roger Granada, **Juarez Monteiro**, Nathan Gavenski, Felipe Meneguzzi. *MICAI 2020*, 2020.
- Improving Action Recognition using Temporal Regions. Roger Granada, João Paulo Aires, **Juarez Monteiro**, Felipe Meneguzzi, Rodrigo C. Barros. *JIDM 2018*, 2018.
- Fusing Scene Context to Improve Object Recognition. Leandro Silva, Roger Granada, **Juarez Monteiro**, Duncan Ruiz. *JIDM 2018*, 2018.
- Norm Conflict Identification using Vector Space Offsets. João Paulo Aires, Roger Granada, **Juarez Monteiro**, Felipe Meneguzzi. *IJCNN 2018*, 2018.
- Evaluating the Feasibility of Deep Learning for Action Recognition in Small Datasets. **Juarez Monteiro**, Roger Granada, João Aires, Rodrigo C. Barros. *IJCNN 2018*, 2018.
- Beating Bomberman with Artificial Intelligence. **Juarez Monteiro**, Roger Granada, Rafael Pinto, Rodrigo C. Barros. *ENIAC 2018*, 2018.
- Using Scene Context to Improve Action Recognition. **Juarez Monteiro**, Roger Granada, Felipe Meneguzzi, Rodrigo C. Barros. *CIARP 2018*, 2018.
- Hybrid Activity and Plan Recognition for Video Streams. Roger Granada, Ramon F. Pereira, **Juarez Monteiro**, Rodrigo C. Barros, Duncan Ruiz, Felipe Meneguzzi. *PAIR-AAAI 2017*, 2017.

- A Deep Neural Architecture for Kitchen Activity Recognition. Roger Granada, **Juarez Monteiro**, Rodrigo C. Barros, Felipe Meneguzzi. *FLAIRS 2017*, 2017.
- Virtual Guide Dog: An Application to Support Visually-Impaired People through Deep Convolutional Neural Networks. **Juarez Monteiro**, João Paulo Aires, Roger Granada, Rodrigo C. Barros, Felipe Meneguzzi. *IJCNN 2017*, 2017.
- Deep Neural Networks for Kitchen Activity Recognition. **Juarez Monteiro**, Roger Granada, Rodrigo C. Barros, Felipe Meneguzzi. *IJCNN 2017*, 2017.
- Deep Neural Networks for Handwritten Chinese Character Recognition. Renan Maidana, **Juarez Monteiro**, Roger Granada, Alexandre Amory, Rodrigo C. Barros. *BRACIS 2017*, 2017.